# DG INTERN HUB

## SOC ANALYST INTERNSHIP

### WEEK 2 — TRYHACKME ROOM WRITEUPS

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**Domain:** Cybersecurity / SOC Analyst

**Platform:** TryHackMe

**Submission Date:** 14 October 2026

# ROOM 1: NETWORK TRAFFIC BASICS

## Learning Objective

The objective of this room was to understand network traffic, common protocols, and how suspicious network activities can help identify possible security threats.

## 1. Introduction

In this room, I learned how devices communicate with each other over a network and how network traffic can be analysed to identify unusual activities. I understood that network traffic analysis is an important skill for SOC analysts because it helps them investigate suspicious connections and possible attacks.

## 2. Key Concepts Learned

Network Traffic Analysis: It involves examining data moving between devices to understand communication and identify unusual behaviour.
DNS Tunnelling: Attackers may misuse DNS queries and responses to transfer hidden data or communicate with command-and-control servers.
HTTP and TLS: HTTP is used for communication between web browsers and web servers. TLS helps protect communication by encrypting data.
TCP Sequence Numbers: TCP uses sequence numbers to keep track of transmitted data and maintain reliable communication.
Packet Fragmentation: Packets can be divided into smaller fragments during transmission. Attackers may misuse fragmentation to make malicious traffic harder to inspect.

SMB and Kerberos: SMB is commonly used for file and resource sharing. Kerberos is an authentication protocol that helps verify identities.

## 3. Practical Learning

During this room, I answered questions related to network traffic, protocols, packet behaviour, and suspicious communication patterns. I learned how unusual DNS queries, unexpected network connections, and attempts to evade security monitoring may indicate suspicious activity.

## 4. SOC Analyst Perspective

A SOC analyst needs to understand normal network behaviour to identify suspicious activity. Unusual DNS requests, unexpected connections, and abnormal traffic patterns may require further investigation.
Network traffic analysis helps analysts investigate alerts and identify possible indicators of compromise.

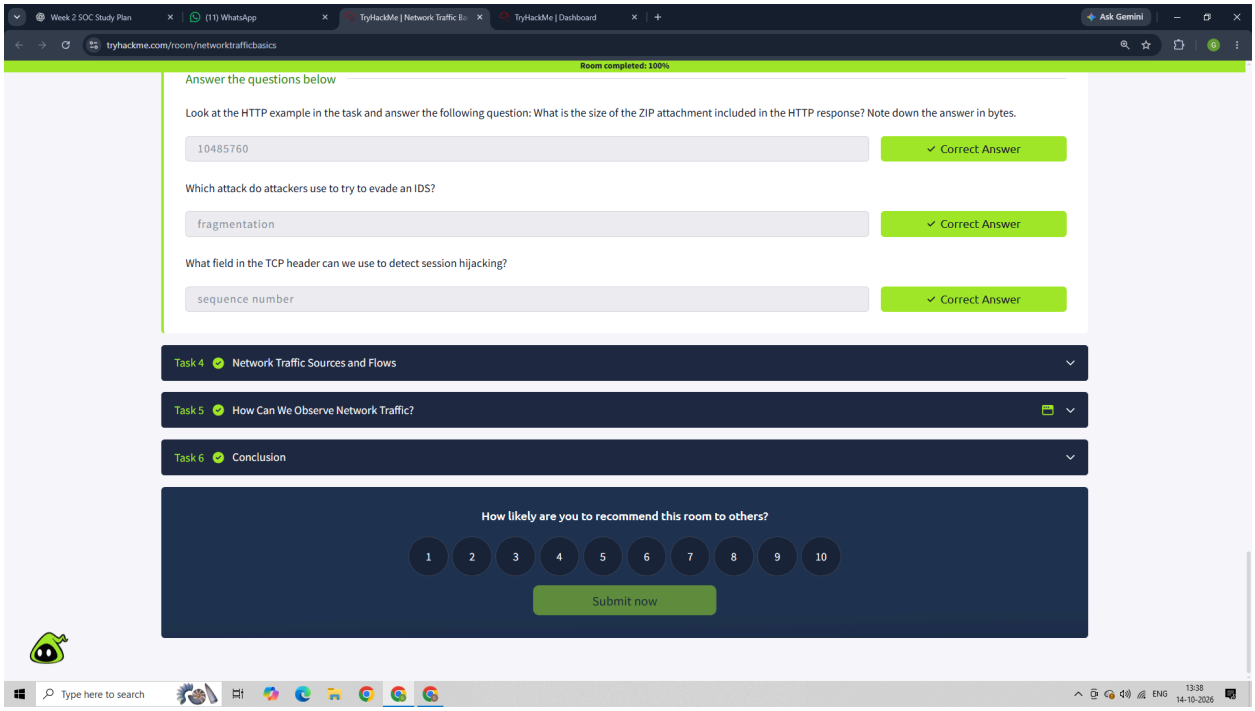
## 5. What I Learned

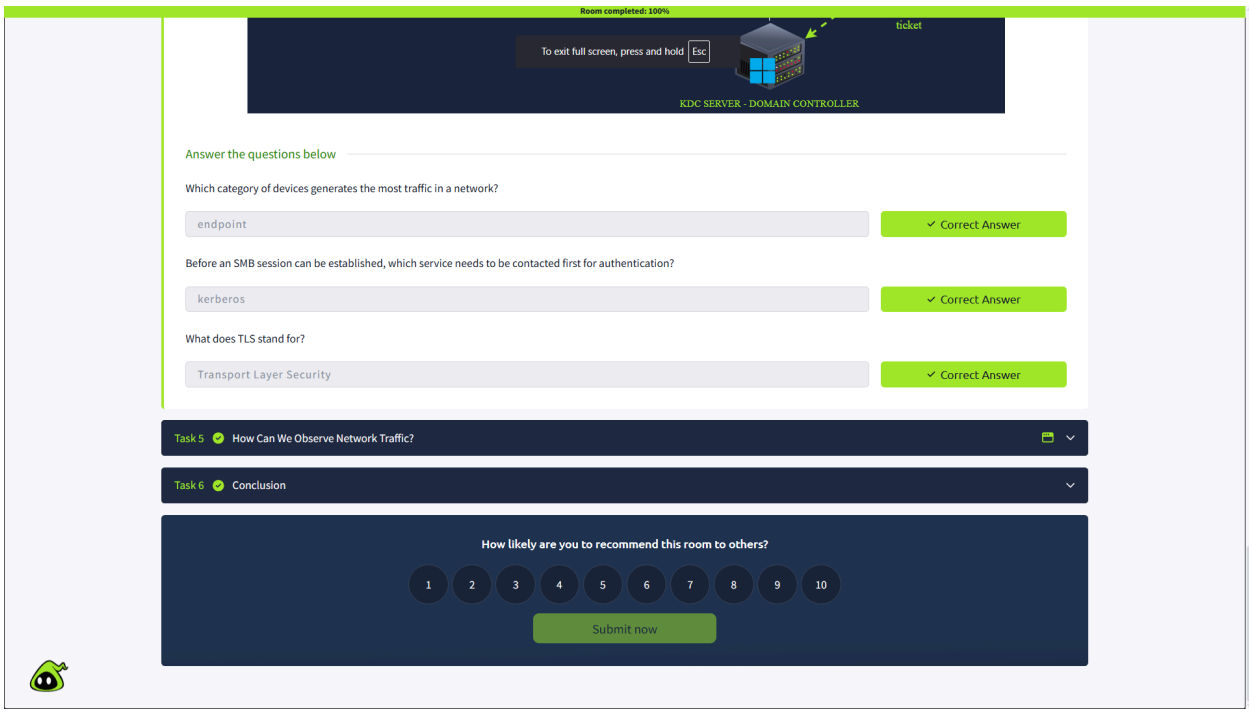
- I understood why network traffic analysis is important in cybersecurity.
- I learned about DNS, HTTP, TLS, TCP, SMB, and Kerberos.
- I understood that attackers can misuse legitimate network protocols.
- I learned how network analysis supports threat detection and SOC investigations.

# 6. Conclusion

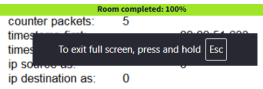
This room helped me improve my understanding of network communication and traffic analysis. I learned that analysing network activity is useful for identifying suspicious behaviour and supporting security investigations.

# 7. Evidence


Week 2 SOC Study Plan
(11) WhatsApp
TryHackMe | Network Traffic Ba
TryHackMe | Dashboard
Ask Gemini
tryhackme.com/room/networktrafficbasics
Room completed: 100%
Answer the questions below
Look at the HTTP example in the task and answer the following question: What is the size of the ZIP attachment included in the HTTP response? Note down the answer in bytes.
10485760
Correct Answer
Which attack do attackers use to try to evade an IDS?
fragmentation
Correct Answer
What field in the TCP header can we use to detect session hijacking?
sequence number
Correct Answer
Task 4 Network Traffic Sources and Flows
Task 5 How Can We Observe Network Traffic?
Task 6 Conclusion
How likely are you to recommend this room to others?
1 2 3 4 5 6 7 8 9 10
Submit now
Type here to search
ENG
13:38
14-10-2026


Room completed: 100%
To exit full screen, press and hold Esc
ticket
KDC SERVER - DOMAIN CONTROLLER
Answer the questions below
Which category of devices generates the most traffic in a network?
endpoint
Correct Answer
Before an SMB session can be established, which service needs to be contacted first for authentication?
kerberos
Correct Answer
What does TLS stand for?
Transport Layer Security
Correct Answer
Task 5 How Can We Observe Network Traffic?
Task 6 Conclusion
How likely are you to recommend this room to others?
1 2 3 4 5 6 7 8 9 10
Submit now

Room completed: 100%




**The Internet Protocol Flow Information Export protocol (IPFIX)** can be considered as the successor to NetFlow. NetFlow was initially a proprietary protocol from Cisco. This means that the protocol was designed for Cisco systems only. Only from NetFlow v9 on did Cisco include templating, so other vendors could adapt it to their devices. In collaboration with Cisco and other vendors, the IETF created IPFIX and released it as a vendor-neutral standard. It offers features similar to NetFlow, but includes more flexibility in configuring which fields to capture.

**To implement NetFlow or IPFIX**, we don't need a whole new set of infrastructure or dedicated servers. Most vendors implement these protocols standard in their devices. We just have to enable and configure the protocol and have a place to send the metadata. You don't need a dedicated server for collecting this data; many NGFWs, IPS, and IDS have an implementation to collect and analyze flow data.

Answer the questions below

What is the flag found in the HTTP traffic in scenario 1? The flag has the format THM{}.

THM{FoundTheMalware} ✓ Correct Answer

What is the flag found in the DNS traffic in scenario 2? The flag has the format THM{}.

THM{C2CommandFound} ✓ Correct Answer

Task 6 ✓ Conclusion

How likely are you to recommend this room to others?

1 2 3 4 5 6 7 8 9 10

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Room completed: 100%

Below are two more scenarios that illustrate the importance of network traffic analysis:

- Based on the logs for an end-user system, the system began to deviate from its normal behavior around 4 PM UTC. Analyzing the network traffic going to and from this system, we found a suspicious HTTP request and were able to extract a suspicious ZIP-file
- We received an alert that an end-user system is sending many DNS requests in comparison to baseline of the network. After inspecting the DNS requests, we discovered that data was being exfiltrated using a technique called DNS tunneling

Now that we know **why** we need network traffic analysis, let's continue with the next task to discover **what** exactly we can monitor.

Answer the questions below

What is the name of the technique used to smuggle C2 commands via DNS?

DNS tunneling ✓ Correct Answer

Task 3 ✓ What Network Traffic Can We Observe?

Task 4 ✓ Network Traffic Sources and Flows

Task 5 ✓ How Can We Observe Network Traffic?

Task 6 ✓ Conclusion

How likely are you to recommend this room to others?

1 2 3 4 5 6 7 8 9 10

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